

Aleena Varghese

AI Engineer • Data Scientist

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Professional Summary

AI Engineer with hands-on internship experience at Lightrains Technolabs, building production-grade Generative AI, RAG, LLM, and Computer Vision applications. M.Sc. Computer Science (Data Analytics) graduate skilled in Python, FastAPI, Django, React, Next.js, TensorFlow, OpenAI API, LightRAG, MinerU, and Vision Language Models, with a strong foundation in designing intelligent, scalable AI solutions. Seeking opportunities as an AI Engineer, Machine Learning Engineer, or Data Scientist.

Experience

Lightrains Technolabs

March 2026 – May 2026

AI Engineer Intern

- Engineered a production-grade Multimodal Diagnostic Assistant using FastAPI, Next.js, LightRAG, and PydanticAI, orchestrating an 8-stage AI diagnostic pipeline for accurate troubleshooting.
- Built an advanced RAG pipeline integrating knowledge graph retrieval, vector search, MinerU document parsing, and intent-aware retrieval to improve technical information extraction.
- Implemented Vision-Language Model (VLM)–based visual grounding, semantic image indexing, and reranking to deliver context-aware, high-precision technical diagrams.
- Designed robust AI safety guardrails, contextual memory, confidence validation, and escalation logic to ensure reliable, explainable, and user-safe diagnostic recommendations.

Projects

HR Assistant System | *React, TypeScript, Django, Python*

- Developed an AI-powered HR assistant using Django, React, TypeScript, and Llama 3.3 to automate payroll, leave, policy, and document queries.
- Designed a secure 9-stage AI orchestration pipeline with intent classification, policy validation, context management, and audit logging for accurate responses.
- Built interactive chat interfaces and integrated AI/LLM workflows for contextual HR support and policy enforcement
- Implemented Zero-Hallucination AI with role-based access, deterministic computations, and database-grounded responses to ensure secure and reliable HR assistance.

Age, Gender, and Emotion Prediction | *Python, CNN, TensorFlow/Keras, OpenCV*

- Developed CNN-based deep learning models using TensorFlow/Keras to predict age, gender, and facial emotions from images using the UTKFace and FER2013 datasets.
- Implemented end-to-end training, evaluation, and inference pipelines with OpenCV, NumPy, Pandas, and Scikit-learn, achieving robust facial attribute recognition.

SignBot – Sign Language Recognition | *Python, OpenCV, SVM*

- Developed a real-time Sign Language Recognition system using OpenAI CLIP and an SVM classifier to convert hand gestures into text with high-accuracy predictions.
- Integrated Google Gemini AI and Text-to-Speech (TTS) to generate intelligent, context-aware text and voice responses, enabling accessible communication.
- Built an end-to-end pipeline for dataset collection, augmentation, feature extraction, model training, and real-time inference using Python, OpenCV, Streamlit, and Scikit-learn.

Education

Digital University Kerala	2024 – 2026
M.Sc. Computer Science (Data Analytics) CGPA: 9.12	Thiruvananthapuram, Kerala
Mar Ivanios College (Autonomous)	2021 – 2024
B.Sc. Mathematics CGPA: 7.71	Thiruvananthapuram, Kerala

Technical Skills

Programming Languages: Python, SQL, JavaScript

AI & Machine Learning: Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, Natural Language Processing (NLP), CNN, SVM

Frameworks & Libraries: FastAPI, Django, React.js, Next.js, Streamlit, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib

GenAI & AI Tools: OpenAI API, Google Gemini AI, Groq Llama 3.3, LightRAG, MinerU, CLIP, Vision Language Models (VLMs), PydanticAI

Databases & Storage: SQLite, Vector Databases, Knowledge Graphs

Relevant Certifications

- **Data Analytics Track** – Google Cloud Career Launchpad
- **Computing Foundations Track** – Google Cloud Career Launchpad
- **Data Analytics Job Simulation** – Deloitte (Applied Analytics, Problem-Solving)
- **Data Visualization** – Tata Group
- **Python Data Structures** – University of Michigan (Coursera)
- **Introduction to Large Language Models** – Google Cloud